

MOHAMED FAWZI YOUSEF MOHAMED

AI Solutions Architect & Full-Stack Engineer (Digital Surgeon)

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| GitHub: <https://github.com/WD-FAWZI> | Portfolio: <https://mf-portfolio-v5-final.vercel.app> |

PROFESSIONAL SUMMARY

A Medical Doctor (MBBS) transitioned into a Software Engineer and AI Solutions Architect. Highly driven "learnaholic" leveraging a rigorous clinical diagnostic mindset to engineer secure, scalable, and automated digital systems. Specializes in modern frontend ecosystems (React, TypeScript) and Artificial Intelligence (Local LLMs, Voice Agents). Proven ability to independently architect end-to-end applications, solving complex architectural bottlenecks from database schema to UI performance.

TECHNICAL SKILLS

Frontend: React, Next.js, TypeScript, Vite, Tailwind CSS, Three.js, Monorepo Architecture.

Backend & Infra: Node.js, Python, FastAPI, Supabase, Firebase, Vercel, RESTful APIs.

AI & Architecture: Local LLMs (Ollama), Voice AI Agents (Vapi, ElevenLabs), Prompt Caching, Multi-Agent Workflows, Brain-Computer Interfaces (BCI).

Core Competencies: Clean Architecture, Type Narrowing, UI/UX Product Design, Medical Data Systems.

ENGINEERING PROJECTS & EXPERIENCE

Itminan Health (December 2025) | Product Designer & Full-Stack Developer

- Architected a full-stack diabetic care platform designed to bridge the gap between patients and healthcare providers.

- Engineered the frontend using React and TypeScript, ensuring strict type safety and a responsive, highly accessible user interface.
- Optimized clinical review processes by centralizing months of scattered patient medical records into actionable, real-time data visual insights.
- Designed the database schema to securely handle sensitive medical data, significantly reducing data retrieval latency.

H.I.M (Voice AI Conversational Agent) | AI Solutions Architect, February 2026

- Developed a low-latency, real-time voice AI agent from the ground up, tailored to handle specific dialect recognition.
- Orchestrated complex external API integrations, combining Vapi and ElevenLabs to achieve human-like conversational fluidity.
- Addressed and resolved critical challenges in real-time audio processing, reducing response latency and ensuring continuous state management during active voice sessions.

AI-Driven EdTech Assessment Tool | AI & Full-Stack Developer | Completed March 2026

- Prototyped and deployed an educational assessment platform capable of generating accurate, context-aware medical questionnaires.
- Implemented advanced Prompt Caching techniques and dynamic model switching to drastically reduce API costs and improve response times.
- Executed successful User Acceptance Testing (UAT), receiving high praise for the tool's precision and speed from initial users.

HACKATHONS & OPEN SOURCE

Mistral AI Hackathon (Feb 28 - Mar 1, 2026) — Project "ER Zero Latency" | Solo AI Solutions Architect & Medical Subject Matter Expert (MBBS)

- **Engineered** a pioneering, zero-latency Multimodal AI co-pilot to **eliminate** Emergency Room triage bottlenecks by transforming passive CCTV surveillance into an active, predictive monitoring system.
- **Leveraged** deep medical domain expertise (MBBS) to **program** the computer vision pipeline to **detect** critical biomechanical distress signals (e.g., Levine's sign) and **correlate** this real-time data with historical EHRs to **predict** Acute Coronary Syndrome (ACS).
- **Architected** and **deployed** a sophisticated **4-Engine Synchronized Orchestration** utilizing advanced Asynchronous Multi-threading to guarantee non-blocking, deterministic performance.

- **Invented** an event-driven 'Low-Power Standby' logic (Eco-Mode) that aggressively **conserved** compute resources, **reducing** GPU consumption by **85%** and ensuring the system's economic viability for mass deployment.
- **Achieved** a critical performance standard of **sub-100ms** response time from visual trigger to final diagnostic alert.
- **Tech Stack:** Python, Mistral AI, MediaPipe, DeepFace, OpenCV.

EDUCATION

Bachelor of Medicine, Bachelor of Surgery (MBBS) - May 2025

Advanced skills in AI and software engineering were rapidly self-taught and executed in the months immediately following medical graduation.

[Ibn Sina University (ISU)]

- Merged complex medical sciences with a self-taught, rigorous curriculum in software engineering and artificial intelligence.